

Q.What is cross staff?list down types of cross staff. describe the construction and uses of a open cross with a sketch

Cross staff is a navigational instrument used to measure the altitude of celestial bodies above the horizon. It was widely used in the past for navigation purposes before the development of more accurate instruments.

There are two main types of cross staff:

Open cross staff: This type of cross staff has a "V" shaped frame with two arms that slide along the frame. One arm has a vertical sighting vane and the other has a horizontal sighting vane.

Closed cross staff: This type of cross staff has a solid frame with a movable vane that slides along a groove in the frame.

Open cross staff construction and uses:

An open cross staff is made up of four main parts: the frame, the two sliding arms, and the vanes. The frame is usually made of wood or metal and is shaped like a "V". The two sliding arms fit into the frame and can be moved up and down. The vertical arm has a sighting vane at the top, and the horizontal arm has a sighting vane at the end.

To use an open cross staff, the navigator would hold the instrument vertically and look through the sighting vanes to line up the celestial body with the horizon. The navigator would then move the arms until the two sighting vanes are aligned with the celestial body and the horizon. The altitude of the celestial body could then be read off a scale on the frame.

The open cross staff was often used by sailors to determine their latitude at sea by measuring the altitude of the sun or stars. However, it was not very accurate and has been largely replaced by more modern navigational instruments.

Sighting vane on top of vertical arm

Sighting vane at end of horizontal arm

Note: The "Sighting vanes" refer to the parts of the arms through which the navigator looks to align the instrument with the celestial body and horizon.

Q.Explain curve ranging, definition of curve, objectives behind provision of curves, constructions survey for waterways

Ans: Curve ranging is a process in surveying that involves determining the location and shape of curves along a path, such as a roadway or waterway. In the case of waterways, curve ranging is often used to determine the location of bends, curves, and turns in the watercourse.

In surveying, a curve is a section of a path or line that is not straight, but instead has a continuously changing direction. Curves are typically defined by their radius, which is the distance from the center of the curve to the center of the circle that best approximates the curve. The degree of curvature of a curve is measured in terms of the central angle of the arc that makes up the curve.

The objectives behind:- the provision of curves in waterway construction surveys are to ensure that the waterway is properly aligned and shaped to maximize its navigability and safety. This includes minimizing sharp turns or bends, ensuring that the waterway is wide enough to accommodate vessel traffic, and ensuring that there are no obstructions or hazards in the path of the waterway.

During a waterway construction survey:- curve ranging is typically conducted using specialized surveying tools such as theodolites, total stations, and GPS receivers. Surveyors use these tools to measure the location and angle of the curve at regular intervals along the waterway. This information is then used to create a map or plan of the waterway that shows the location, degree, and radius of each curve, as well as any other relevant features such as obstructions, hazards, or navigational aids. This information is critical for the safe and efficient navigation of the waterway.

Q. Define the term contour line. explain the characteristics of contour lines with diagram.

Ans : A contour line is a line on a map or a graph that connects points of equal elevation or altitude. In other words, it represents a line that connects all the points on a given surface that are at the same height above sea level.

Here are some characteristics of contour lines:

#Contour lines never cross each other: This is because they represent points of equal elevation, and if they were to cross, that would mean there are two different elevations at that point, which is not possible.

#Contour lines can be close together or far apart: When contour lines are close together, it means the elevation change is steep, while when they are far apart, it means the elevation change is gradual.

#Contour lines can be used to determine the shape of the land: By looking at the contour lines, you can tell if the land is flat, sloping, or steep.

#Contour lines are often labeled with their elevation: This allows you to know the exact elevation of a particular point on the map.

Contour Lines Diagram

In the diagram, the contour lines are represented by the brown lines. As you can see, the lines are close together where the terrain is steep, and they are far apart where the terrain is flat or gently sloping. The labeled numbers on the lines show the elevation in feet above sea level. By using this map, you can determine the elevation of any point on the terrain, as well as its shape and steepness.

Q.Explain the transit theodolite with diagram

Ans A transit theodolite is a surveying instrument used to measure horizontal and vertical angles in order to determine the relative positions of points on the earth's surface. It consists of a telescope mounted on a vertical axis that can be rotated horizontally around a horizontal axis. The telescope can also be tilted vertically to measure angles of elevation and depression.

The main components of a transit theodolite are:

Telescope: This is the main part of the transit theodolite. It consists of a lens at the front and an eyepiece at the back. The telescope can be rotated horizontally around a vertical axis and tilted vertically to measure angles of elevation and depression.

Base: This is the part of the instrument that supports the telescope and the horizontal circle. It is usually mounted on a tripod.

Horizontal Circle: This is a circular disk that is attached to the base. It is marked with degrees, minutes, and seconds, and is used to measure horizontal angles.

Vernier: This is a device that is used to read the angles on the horizontal circle with greater precision than is possible with the naked eye.

Leveling Screws: These are used to level the instrument before taking measurements.

Vertical Circle: This is a circular disk that is attached to the telescope. It is marked with degrees, minutes, and seconds, and is used to measure vertical angles.

Transit: This is a mechanism that allows the telescope to be flipped over so that it points in the opposite direction. This is used to measure angles in a complete circle.

Overall, the transit theodolite is a versatile and precise instrument that is widely used in surveying and mapping applications.

Q.Bring out the difference between open and closed traverse .

Ans:Open and closed traverse are two methods of surveying land or terrain. The key differences between the two methods are as follows:

Definition: An open traverse is a survey where the starting and ending points are not the same, while a closed traverse is a survey where the starting and ending points are the same.

Objective: An open traverse is generally used for exploring a new area or when it is difficult or impossible to return to the starting point. In contrast, a closed traverse is used when the goal is to obtain a more accurate measurement of an area or to check the accuracy of existing measurements.

Accuracy: A closed traverse is generally more accurate than an open traverse because it eliminates the cumulative errors that may occur in an open traverse due to inaccuracies in individual measurements.

Equipment: The equipment used in an open traverse is generally simpler than that used in a closed traverse. Open traverses often use a compass, measuring tape, and stadia rod, while closed traverses may use more advanced equipment such as a theodolite or total station.

Time: A closed traverse generally takes longer to complete than an open traverse because it requires more measurements to be taken and more calculations to be made.

Surveying method: In an open traverse, the surveyor may take any route necessary to reach the endpoint, while in a closed traverse, the surveyor must follow a predetermined path and return to the starting point.

In summary, the main difference between an open and a closed traverse is that an open traverse has a starting and ending point that are not the same, while a closed traverse has a starting and ending point that are the same. The choice between the two methods depends on the purpose of the survey, the accuracy required, and the terrain being surveyed.

Q.Explain the construction and uses of surveyors compass.

Ans:A surveyor's compass, also known as a circumferentor, is a tool used in land surveying to measure angles and bearings. The instrument consists of a circular compass housing, which contains a magnetic needle that aligns with the Earth's magnetic field, and a sighting device called an alidade.

#The construction of a surveyor's compass typically involves a circular metal or brass housing that contains a compass needle that is suspended on a pivot. The needle is balanced on a jeweled bearing or agate cap, which helps reduce friction and allows for smoother rotation. The housing is typically marked with degrees or other units of measurement, and the needle can be locked in place for reading and recording measurements.

#The alidade is attached to the top of the compass housing and consists of a sighting device, usually a telescope or sight vane, that allows the user to align the compass with a target object or landmark. The alidade also typically has a vertical angle measurement device, which can be used to measure the height of objects or features on the ground.

#Surveyor's compasses are used in land surveying to measure angles and bearings, which are used to create maps, establish boundaries, and determine property lines. The instrument can be used to determine the direction of a line, the angle between two lines, or the position of a point relative to a known reference point. Surveyor's compasses are also useful in construction and engineering projects,

Q. Define orientation in PT surveying, enlist methods of orientation of a plane table and explain in brief any one method of orienting a plane table

Ans: Orientation in plane table surveying refers to the process of setting up the plane table so that its plane is parallel to the surface of the earth and the table is oriented in the desired direction. This is necessary to ensure accurate measurements and plotting of points on the table.

There are several methods for orienting a plane table, including the following:

@The magnetic needle method

@The trough compass method

@The solar compass method

@The resection method

One of the most commonly used methods for orienting a plane table is the magnetic needle method. In this method, a magnetic needle is mounted at the center of the table, which is free to rotate. The plane table is placed on a tripod and leveled using the leveling screws. The magnetic needle is then brought into alignment with the magnetic meridian by adjusting the table until the needle aligns with the North-South direction indicated by a compass.

Once the magnetic needle is aligned with the meridian, the plane table is rotated until the needle points to a point of known azimuth, such as the sun, a star, or a known landmark. This process is known as "sighting" and is done by rotating the table until the needle points to the desired point. The orientation is then considered complete and the surveying can begin.

Overall, the magnetic needle method is relatively simple and quick to use, making it a popular choice for orienting a plane table in the field.

Q.Enumerate the accessories required with a plane table and state use of each of it

Ans : A plane table is a device used in surveying to create an accurate map of an area. The accessories required with a plane table include:

Alidade: An alidade is a straightedge attached to a sighting device that is used to measure angles. It helps to measure the horizontal and vertical angles between two points on the ground.

Tripod Stand: A tripod stand is a three-legged support used to keep the plane table stable and level. It helps to keep the plane table steady during the surveying process.

Plumbing Fork: A plumbing fork is a device used to center the plane table on the tripod stand. It helps to ensure that the plane table is level.

Spirit Level: A spirit level is used to ensure that the plane table is level. It is placed on the table to ensure that the table is perfectly horizontal.

Ranging Rods: Ranging rods are poles that are used to mark points on the ground. They are used to measure distances between points on the map.

Chain or Tape: A chain or tape is used to measure distances between points on the ground. It is used to accurately measure the length of the sides of the triangles formed by the surveying process.

Drawing Materials: Drawing materials like a ruler, protractor, pencil, and eraser are used to create the map of the area being surveyed. The ruler is used to draw straight lines, while the protractor is used to measure angles. The pencil is used to draw the map, and the eraser is used to correct any mistakes.

Each of these accessories is important in the surveying process with a plane table. They help to ensure that the map created is accurate and reliable.

Q.Explain the methods of measurement of horizontal angle by method of repetition with a theodolite.

Ans The method of repetition is a common technique used to measure horizontal angles using a theodolite. Here are the steps to follow:

1.Set up the theodolite at a convenient position and level it properly using the leveling screws.

2.Point the telescope of the theodolite towards the first point you wish to measure the horizontal angle to. This point is called the back station.

3.Using the horizontal clamp and tangent screw, turn the telescope until the vernier reads zero on the horizontal circle.

4.Now, turn the telescope towards the second point which is called the forward station.

5.Note the reading on the horizontal circle, which gives the angular measurement between the back station and the forward station.

6.To increase the accuracy of the measurement, repeat the process several times, taking note of the readings on the horizontal circle each time.

7.Calculate the mean of the observed readings to get a more accurate value for the horizontal angle.

8.It is also important to make sure that the theodolite remains stable throughout the measurement process, and any disturbance is avoided.

Additionally, it is recommended to use a target on the forward station to aid in the accuracy of the measurement.

By repeating the process several times and taking the mean, errors due to instrument or observational inaccuracies can be minimized, resulting in a more precise measurement of the horizontal angle.

Q.Prepare brief note on preliminary survey for a road/railway .

Ans : A preliminary survey for a road or railway is an important step in the construction of a new transportation infrastructure. It is typically conducted to gather information about the proposed route and to identify any potential issues or obstacles that may need to be addressed before construction can begin.

The preliminary survey may involve a variety of tasks, including:

Route identification: The first step in the survey is to identify potential routes for the road or railway. This may involve analyzing maps and satellite images, conducting site visits, and consulting with local authorities.

Topographic survey: Once the route has been identified, a topographic survey is conducted to gather information about the terrain and existing features such as rivers, hills, and other natural obstacles.

Soil investigation: A soil investigation is carried out to determine the soil composition, stability, and load-bearing capacity of the ground in the proposed route.

Environmental impact assessment: An environmental impact assessment is conducted to determine the potential impact of the road or railway on the surrounding environment, including flora and fauna, air and water quality, and noise levels.

Cost estimation: Based on the information gathered during the preliminary survey, a rough estimate of the construction cost is prepared.

The results of the preliminary survey are used to inform the design of the road or railway, and to identify any additional data or investigations that may be required before construction can begin. It is an essential step in the process of building a new transportation infrastructure and can help to ensure that the project is completed on time, within budget, and with minimal impact on the environment.

Q. List down the methods of plane tabling. Explain procedure of any two methods with neat figures

Ans: Plane tabling is a surveying method that involves creating a map or plan of an area using a plane table, which is a device that consists of a flat surface mounted on a tripod. The following are the methods of plane tabling:

Radiation Method: In this method, the plane table is set up at a known point, and a single ray or line is drawn from the table to each survey station. The direction and distance of each station are then measured using a compass and a tape measure, respectively.

Intersection Method: In this method, the plane table is set up at a known point, and two or more lines are drawn from the table to each survey station. The angles between the lines are measured using a protractor, and the distance from the table to each station is measured using a tape measure.

Procedure of the Radiation Method:

Set up the plane table at a known point, such as a benchmark or a previously surveyed point. Level the table using the bubble level on the tripod.

Orient the table so that the compass needle points to magnetic north or true north, depending on the type of compass being used.

Choose a survey station that can be seen from the table. Take a magnetic bearing from the table to the station using the compass. Measure the distance from the table to the station using a tape measure.

Mark the position of the station on the table using a pencil or pin.

Repeat steps 3-4 for each survey station.

Connect the marks on the table using a straightedge or ruler to create a map or plan of the area.

Procedure of the Intersection Method:

Set up the plane table at a known point, such as a benchmark or a previously surveyed point. Level the table using the bubble level on the tripod.

Orient the table so that the compass needle points to magnetic north or true north, depending on the type of compass being used.

Choose two or more survey stations that can be seen from the table. Take magnetic bearings from the table to each station using the compass. Measure the distance from the table to each station using a tape measure.

Mark the position of each station on the table using a pencil or pin.

Draw a line from the table to each station using a straightedge or ruler.

Measure the angles between the lines using a protractor.

Repeat steps 3-6 for each set of survey stations.

Connect the lines on the table using a straightedge or ruler to create a map or plan of the area.